Chapter 1: Introduction to Causal Analysis

Inference and Intervention | BK Advisors

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Learning Objectives

After this chapter, you will be able to:

- Distinguish between **situational assessment** ("What is happening?") and **managerial intervention** ("What should I do?")
- Explain why **correlation does not imply causation** using real-world health examples
- Identify three classic traps in causal thinking: **confounding**, **reverse causation**, and **selection bias**
- Describe the basic building blocks of a **causal model** — nodes, arrows, and directed acyclic graphs (DAGs)
- Explain why causal reasoning matters for designing health programs that actually save lives

Chapter Overview

Situational Assessment → Managerial Intervention → Correlation ≠ Causation → Causal Model Framework → MNH Application

The big idea: When resources are scarce — and they always are — you need to know *what causes what* before you can decide *what to do*. This chapter introduces the thinking tools that make that possible.

Situational Assessment

What is happening? What are the causes?

Here's the Puzzle

Ethiopia has one of the largest community health worker programs in Africa — over **40,000 Health Extension Workers** deployed across the country.

But here is something strange:

> **The regions where the most Health Extension Workers are deployed tend to have the highest mortality rates.**

Does deploying health workers *cause* deaths?

Obviously not. But if you just looked at the numbers — workers deployed vs. mortality rates — that is exactly what the data would seem to say.

> **This is the kind of puzzle that causal thinking solves.** The data is not lying. But it is not telling the whole story either. By the end of this chapter, you will be able to explain exactly what is going on.

The Manager's Two Questions

Every decision-maker faces two fundamental questions:

Question 1: What is happening?

Situational Assessment — understanding the current state by gathering evidence and reasoning about causes.

"Why are so many mothers still dying in Amhara region despite having more health workers than anywhere else?"

Question 2: What should I do?

Managerial Intervention — choosing which levers to pull to change outcomes.

"Should we train more midwives, buy more equipment, or fix the data system?"

Key insight: You cannot answer Question 2 well until you have answered Question 1 well. Acting without understanding causes is like prescribing medicine without a diagnosis.

What Is Situational Assessment?

Situational Assessment: Using evidence to reason about the current state of a system — figuring out which variables are in which states, and *why*.

Think of it this way:

- A **doctor** examining a patient: What symptoms are present? What do they point to?
- A **mechanic** diagnosing a car: The engine is overheating — but is it the coolant, the fan, or a cracked gasket?
- A **health program manager** in Ethiopia: Neonatal mortality is not falling in Somali region — is it a workforce problem, a supply chain problem, or something else entirely?

In each case, the person is trying to **figure out what is causing what** before deciding how to act. Causal models give us a systematic

Ethiopia's Health Challenge: The Numbers

Here is the situation Ethiopia faces, based on publicly available data:

Indicator	Ethiopia	Source
Maternal Mortality Ratio	~267 per 100,000 live births	WHO, 2020
Neonatal Mortality Rate	~28 per 1,000 live births	UNICEF, 2021
Health Extension Workers	40,000+ deployed nationally	Ethiopia MoH
Antenatal care coverage (at least 1 visit)	~74%	DHS, 2019
Leading causes of maternal death	Hemorrhage, hypertension, sepsis, obstructed labor	WHO

Ethiopia has made enormous progress — maternal mortality dropped from over 1,000 per 100,000 in the 1990s to about 267 today. But there is still a long way to go, and where to invest next depends on understanding what is *causing* the remaining deaths.

Managerial Intervention

What should I do? Which levers should I pull?

Managers Manage Scarce Resources

Managerial Intervention: A decision about which things to control or change — the choice of *what* to act on, before deciding *how* to act.

- Resources are always limited — money, time, people, equipment
- Every investment involves **trade-offs**: spending on training means *not* spending on equipment
- Choosing wisely requires understanding **what causes what**

Imagine you are running a health program in Ethiopia with a fixed budget. You have to choose how to divide your resources. Here are three options:

The Trade-Off

OPTION A: PEOPLE

Train 5,000 more midwives in emergency obstetric care. Staff every health center with a skilled birth attendant.

Theory: More trained staff → better care → fewer deaths.

OPTION B: PRODUCTS

Buy CPAP machines for newborn breathing support, PPH kits for hemorrhage treatment, and ultrasound devices for every district hospital.

Theory: Better equipment → better treatment → fewer deaths.

OPTION C: SYSTEMS

Build a real-time data dashboard (DHIS2), a referral transport network, and a supply chain tracking system.

Theory: Better information → better decisions → better care → fewer deaths.

Here is the problem: All three sound reasonable. But you cannot do all three equally well. So which one will *cause* the greatest reduction in mortality? That is a causal question — and you cannot answer it with a spreadsheet full of correlations.

Why You Need Causal Thinking to Decide

Consider what happens if you skip the causal reasoning and just look at the data:

WHAT THE DATA MIGHT SHOW

Regions with the **most equipment** have the **lowest mortality**.

Naive conclusion: Buy more equipment!

WHAT IS ACTUALLY GOING ON

Regions with the most equipment are also the **wealthiest, most urban regions** — they would have had lower mortality anyway.

The equipment is not causing the low mortality. Wealth is causing both.

Without causal thinking, data can lead you to exactly the wrong decision.

The rest of this chapter teaches you *why* this happens and *how* to avoid it.

Correlation ≠ Causation

The three traps of causal reasoning

The Ice Cream Problem

Before we go back to health data, here is a classic example everyone can relate to:

EVERYDAY ANALOGY

Observation: On days when ice cream sales are high, drowning deaths are also high.

Naive conclusion: Ice cream causes drowning!

Reality: Hot weather causes *both*. People eat more ice cream in summer. People also swim more in summer. **Summer heat** is the hidden third variable connecting both.

This hidden-third-variable problem has a name: **confounding**. And it is just one of three traps that make people confuse correlation with causation.

Trap 1: Confounding

Confounding occurs when a third variable influences both the supposed cause and the supposed effect, creating a misleading association between them.

MNH EXAMPLE

Observation: Regions in Ethiopia with more Health Extension Workers deployed have *higher* mortality rates.

Naive conclusion: Health workers cause deaths?

Reality: More health workers are deployed **to the sickest regions** — places with the highest disease burden. **Disease burden** is the confounder. It drives both health worker deployment (because that is where the need is) *and* mortality (because that is where disease is worst).

Read the diagram: Disease burden sends arrows to *both* health worker deployment *and* mortality. That shared cause creates a correlation between workers and mortality — even though workers actually *help*.

Trap 2: Reverse Causation

Reverse Causation occurs when the supposed effect actually causes the supposed cause — the arrow points the wrong way.

MNH EXAMPLE

Observation: Countries that receive more international health funding tend to have *higher* mortality rates.

Naive conclusion: Foreign aid increases mortality?

Reality: High mortality **attracts** funding. Donors send money to the countries with the worst health outcomes. The causal arrow runs from mortality to funding, not the other way around.

✗ What people assume:

More Funding → Higher Mortality

✓ What is actually happening:

Higher Mortality → More Funding

Think of it this way: Seeing more firefighters at bigger fires does not mean firefighters cause fires. They are responding to them. Same logic.

Trap 3: Selection Bias

Selection Bias occurs when the data sample is not representative of the population, because which cases end up in your data depends on the very things you are trying to study.

MNH EXAMPLE

Observation: Among hospitals that report data to Ethiopia's DHIS2 system, CPAP machine availability appears unrelated to neonatal survival.

Problem: Only **well-resourced hospitals** consistently report data to DHIS2. These hospitals already have better outcomes. The hospitals where CPAP machines would make the *biggest* difference — small, under-resourced facilities — are missing from the data entirely.

Selection mechanism: Hospital capacity determines both whether the hospital reports data *and* how well its babies survive.

The danger: If you only study the hospitals you can see in the data, you might conclude that equipment does not matter — when in fact it matters enormously for the facilities you *cannot* see.

Three Traps: Summary

Trap	What Goes Wrong	Ethiopia Example
Confounding	A hidden third variable drives both X and Y	Disease burden drives both worker deployment and mortality
Reverse Causation	Y causes X, not X causes Y	Mortality attracts international funding
Selection Bias	Your data sample is not representative	Only well-resourced hospitals report to DHIS2

All three traps create patterns in the data that look like causal effects but are not.

The antidote? Explicit causal modeling — drawing a picture of what causes what *before* you analyze the data.

The Causal Model Framework

Nodes, arrows, and directed graphs

What Is a Causal Model?

Think of it this way: a causal model is just a **diagram** that shows what causes what.

It has three simple ingredients:

- **Nodes** — the things you care about (drawn as circles, rectangles, or other shapes)
- **Arrows** — causal links from one thing to another ("A causes B" is drawn as A → B)
- **Signs** — [+] or [-] telling you the direction of effect (does more of A mean more of B, or less?)

> **The key rule:** An arrow from A to B means that *changing* A can change B. But changing B does *not* change A. Arrows have a direction — and the direction matters.

What Is a DAG?

Directed Acyclic Graph (DAG): A diagram where all arrows point in one direction and there are no loops — you can never follow the arrows from any node back to itself. “Directed” means arrows have a direction. “Acyclic” means no circles. “Graph” is just the math word for a diagram of connected nodes.

Valid DAG (no loops)

A → B → C

You can follow the arrows forward but never get back to where you started.

Not a valid DAG (has a loop)

A → B → C → A

Following the arrows brings you back to A. This is a cycle, not a DAG.

Why does this matter? Loops make it impossible to figure out what causes what — everything causes everything. DAGs force you to be clear and specific about the direction of causation.

A Simple 3-Node Chain

Here is the simplest useful causal model for a health program:

- **Investment → Coverage [+]:** More funding increases the percentage of mothers who receive skilled care
- **Coverage → Mortality [-]:** Higher coverage of life-saving interventions reduces maternal deaths
- This is a **chain** — investment does not directly reduce mortality; it works *through* coverage

> This tiny model already tells us something useful: if investment goes up but coverage does *not* go up, the money is being wasted

But Wait — It Is Not That Simple

The 3-node chain is a good start, but it ignores a lot. What about:

- **Confounders** like disease burden, that affect both where investment goes and how high mortality is?
- **Multiple pathways** — investment might improve training, equipment, *and* data systems, each affecting mortality differently?
- **Quality of care** as a mediator between inputs and outcomes?

That is what the rest of the course is about. In Chapter 2, we will learn the formal rules for building these diagrams. In Chapters 4-6, we will add numbers. In Chapters 7-8, we will use them to make decisions. For now, just understand the basic building blocks: **nodes**, **arrows**, and **direction**.

Preview: A Richer Model

Here is a taste of where we are heading — a more realistic model with multiple pathways:

MNH Application: Ethiopia

Causal thinking applied to real health decisions

Ethiopia's Three-Pillar Approach

Ethiopia's Health Extension Program — one of the most ambitious in Africa — has invested across three pillars:

People	Products	Systems
40,000+ Health Extension Workers	Essential medicines, equipment, supplies	DHIS2, referral networks, supervision

The causal question: How do these three pillars combine to produce better outcomes? Do they work independently, or do they need each other?

Think of it this way: a trained midwife (People) without a PPH kit (Products) cannot stop a hemorrhage. A PPH kit without a trained midwife is just a box on a shelf. And neither one helps if the mother cannot get to the facility (Systems).

People, Products, and Systems are *complements* — they multiply each other's effect.

Ethiopia's Causal Chain

Here is how these pillars connect to outcomes:

Notice: All three pillars flow into Quality of Care. Quality is the bottleneck — the place where everything converges. If quality is poor, no amount of spending on any one pillar will fix the outcomes.

Why the Numbers Mislead

Now let's go back to our opening puzzle. If you looked at raw data across Ethiopia's regions:

WHAT THE DATA SHOWS

Regions with the most Health Extension Workers per capita also have the highest maternal mortality.

Regions with the most equipment also tend to have high mortality.

A naive analyst might say: "These investments are not working!"

WHAT THE CAUSAL MODEL REVEALS

The regions with the highest **disease burden** — think Somali, Afar, Gambella — receive the most resources *because* they have the worst outcomes.

Confounding by disease burden creates a positive correlation between resources and mortality, even when resources are saving lives.

This is exactly Trap 1 (confounding) in action. The solution is not to ignore the data. The solution is to build a causal model that includes disease burden as a confounder, and then analyze the data *through* that model.

ANC Coverage: A Closer Look

Ethiopia's antenatal care (ANC) coverage shows how much progress has been made — and how far there is to go:

- **~74% of pregnant women** attend at least one ANC visit (DHS 2019)
- But only about **43%** complete the recommended four or more visits
- **Urban coverage** is much higher than **rural coverage** — another confounder
- The leading causes of maternal death — hemorrhage (~25%), hypertension (~16%), sepsis (~11%) — are all treatable *if* mothers reach skilled care in time

Here is the causal chain: ANC visits → Early detection of complications → Timely referral → Skilled treatment → Survival. Every link in this chain must work. If *any* one link breaks — a missed visit, a failed referral, an absent midwife, a missing drug — the outcome can be death.

R Workshop: Your First Causal Diagrams

Using dagitty and ggdag in R

Setting Up the R Environment

Before we draw any diagrams, we need to load our tools. The code below loads three R packages that we will use throughout this course.

```r
# Install these packages the first time you use them (remove the # to run):
# install.packages(c("dagitty", "ggdag", "ggplot2"))

# Load the packages into our R session
library(dagitty)    # dagitty lets us define and analyze causal diagrams (DAGs)
library(ggdag)      # ggdag draws pretty pictures of those DAGs
library(ggplot2)    # ggplot2 is the general-purpose plotting library
```

What do these packages do?

- **dagitty** — lets you write down a causal diagram in code and ask questions about it (like “what do I need to control for?”)
- **ggdag** — turns that code into a visual diagram you can look at
- **ggplot2** — the engine that powers the drawings (ggdag uses it behind the scenes)

R: Drawing Our Simple 3-Node Chain

Let's start by drawing the simple Investment → Coverage → Mortality chain we saw earlier. The code below defines three nodes and two arrows, then draws the picture.

```r
# Step 1: Define the DAG
# We list nodes with positions (pos="x,y") and arrows (->)
mnh_simple <- dagitty('dag {
  Investment [pos="0,0"]
  Coverage   [pos="1,0"]
  Mortality  [pos="2,0"]

  Investment -> Coverage
  Coverage -> Mortality
}')

# Step 2: Draw it
# ggdag() takes our DAG definition and produces a plot
# theme_dag() removes the gridlines and axes to keep it clean
ggdag(mnh_simple) +
  theme_dag() +
  labs(title = "Simple MNH Causal Model",
       subtitle = "Investment → Coverage → Mortality")
```

What to notice: The arrows only go left-to-right. Investment affects Coverage, and Coverage affects Mortality. There is no arrow from Investment directly to Mortality — the effect goes *through* Coverage.

R: Adding a Confounder

Now let's add Disease Burden as a confounder — the hidden variable that was creating the misleading correlation between health workers and mortality. This is the confounding diagram from our Ethiopia example.

```r
# Define the confounded model
# Disease_Burden affects BOTH Health_Workers and Mortality
confounded_model <- dagitty('dag {
  Disease_Burden [pos="1,0"]
  Health_Workers [pos="0,1"]
  Mortality      [pos="2,1"]

  Disease_Burden -> Health_Workers
  Disease_Burden -> Mortality
  Health_Workers -> Mortality
}')

# Draw it with color-coding:
# - "exposure" = the thing we think is the cause (Health_Workers)
# - "outcome" = the thing we are measuring (Mortality)
ggdag_status(confounded_model,
             exposure = "Health_Workers",
```

```
              outcome = "Mortality") +
```

Read the picture: Disease Burden sits at the top, sending arrows down to *both* Health Workers and Mortality. That is the fork structure — the classic confounding pattern. Without accounting for Disease Burden, you would wrongly conclude that health workers increase mortality.

R: What Do We Need to Control For?

Here is the powerful part. Once we have drawn the diagram, R can tell us *exactly* which variables we need to control for to get the true causal effect.

```r
# Ask dagitty: "What do I need to control for to estimate the
# causal effect of Health_Workers on Mortality?"
adjustmentSets(confounded_model,
               exposure = "Health_Workers",
               outcome = "Mortality")

# This should return: { Disease_Burden }
# Translation: "Control for Disease Burden, and the confounding goes away."
```

What this means in plain English: If you compare regions that have the *same* disease burden, then the regions with more health workers will have *lower* mortality. The confounding disappears once you account for why workers were deployed where they were.

This is the whole point of the course in one slide. Draw the causal diagram. Ask R what to control for. Analyze the data correctly. Make better decisions. We will build on this foundation in every chapter that follows.

R: Ethiopia's Three-Pillar Model

Let's build the richer model that includes Ethiopia's three-pillar approach — People, Products, and Systems — with Disease Burden as a confounder.

```r
# Define Ethiopia's three-pillar theory of change as a DAG
ethiopia_toc <- dagitty('dag {
  People          [pos="1,0"]
  Products        [pos="1,1"]
  Systems         [pos="1,2"]
  Quality         [pos="2,1"]
  NMR             [pos="3,0.5"]
  MMR             [pos="3,1.5"]
  Disease_Burden  [pos="2,2.5"]

  People   -> Quality
  Products -> Quality
  Systems  -> Quality
  Quality  -> NMR
  Quality  -> MMR
  Disease_Burden -> NMR
  Disease_Burden -> MMR
}')

```

What this model says: All three pillars flow into Quality of Care. Quality then reduces both NMR and MMR. Disease Burden affects the outcomes directly — a region with severe malaria burden will have higher mortality even with good care.

R: Finding All the Paths

We can also ask R to show us every causal pathway from a starting point to an endpoint. This helps us see all the different routes through which an investment could affect mortality.

```r
# What are all the paths from People (workforce investment) to NMR?
people_to_nmr <- paths(ethiopia_toc,
                       from = "People",
                       to = "NMR")

# Print the paths — each one is a different causal route
print(people_to_nmr)

# What variables must we control for to isolate
# the effect of People on NMR?
adjustmentSets(ethiopia_toc,
               exposure = "People",
               outcome = "NMR")
```

Try It: Modify the code above to look at the paths from “Systems” to “MMR” instead. How many routes are there? What do you need to control for?

Key Takeaways

Causal thinking starts with two questions:

"What is happening?" (situational assessment) and "What should I do?" (managerial intervention). You cannot answer the second well without answering the first.

Correlation does not equal causation.

Three traps — confounding, reverse causation, and selection bias — all create misleading patterns in data.

A causal model (DAG)

makes your assumptions visible, testable, and actionable. It tells you what to control for, where the chain breaks, and which interventions will actually work.

Ethiopia's 40,000 Health Extension Workers are not causing deaths.

Disease burden is confounding the relationship. Causal models help us see through the numbers to the truth.

Looking Ahead

Next: Chapter 2 — Qualitative Causal Models

Now that you know *why* we need causal diagrams, we will learn *how* to build them properly:

- Four types of nodes: probabilistic, objective, strategic option, and function
- How to assign [+] and [-] signs to every arrow
- The three fundamental structures: **chains**, **forks**, and **colliders** — and why colliders are the most surprising
- How to determine whether two variables are truly independent (d-separation)

By the end of Chapter 2, you will be able to draw a complete causal diagram for any health program — and use it to make testable predictions about what the data should look like.

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